

Power BI Project Documentation

Integrated Financial Loan Analysis

INDEX:

1. Project Overview
2. Project Objective
3. Data Cleaning And Preprocessing
4. Tools and Technologies Used
5. DAX Measures Used in powerBi
6. Good Loan Analysis Measures
7. Bad Loan Analysis Measures
8. Loan Term Analysis Measures
9. Business Benefits
10. Conclusion

Project Overview

- The Bank Loan Analysis Dashboard is an interactive Power BI solution developed to analyze loan application trends, loan performance, borrower characteristics, and lending patterns.
- The dashboard provides business users with actionable insights into good and bad loans, regional lending distribution, loan purposes, employment profiles, and home ownership categories.

Project Objectives

- Analyze overall loan portfolio performance.
- Identify Good Loans and Bad Loans based on loan status.
- Monitor monthly loan application and funding trends.
- Evaluate regional loan distribution across states.
- Analyze loan applications by employment length and loan term.
- Identify major borrowing reasons through loan purpose analysis.
- Compare lending patterns across home ownership categories.
- Enable interactive navigation using bookmarks and action buttons.

Data Cleaning and Preprocessing

Data cleaning was performed using Python in Jupyter Notebook before loading the dataset into Power BI.

Data Preparation Steps

- Checked missing values using `isnull()`.
- Removed null records where necessary.
- Filled empty values using `fillna()`.
- Removed duplicate records using `drop_duplicates()`.
- Validated data types and column consistency.
- Ensured clean and reliable data for reporting and analysis.

These preprocessing steps improved data quality and ensured accurate dashboard calculations.

Technical Implementation

Data Cleaning and Preprocessing (Python - Jupyter Notebook)

The raw loan dataset was cleaned and transformed using Python in Jupyter Notebook before importing it into Power BI.

Libraries Used

```
import pandas as pd
```

```
import numpy as np
```

Load Dataset

```
df = pd.read_csv("financial_loan.csv")
```

Check Missing Values

```
df.isnull().sum()
```

Fill Missing Values

```
df.fillna(0, inplace=True)
```

or

```
df['column_name'].fillna(df['column_name'].mode()[0], inplace=True)
```

Remove Duplicate Records

```
df.drop_duplicates(inplace=True)
```

Verify Duplicate Removal

```
df.duplicated().sum()
```

Check Dataset Information

```
df.info()
```

Statistical Summary

```
df.describe()
```

Export Cleaned Dataset

```
df.to_excel("financial_loan.xlsx", index=False)
```

Tools and Technologies Used

Tool	Purpose
Python (Jupyter Notebook)	Data Cleaning and Preprocessing
Pandas	Data Transformation
Power BI Desktop	Data Visualization and Dashboard Development
DAX	KPI and Measure Calculations

DAX Measures Used in Power BI

Total Loan Applications

Total Loan Applications =
DISTINCTCOUNT(Financial_Loan[id])

Total Funded Amount

Total Funded Amount =
SUM(Financial_Loan[loan_amount])

Total Amount Received

Total Amount Received =
SUM(Financial_Loan[total_payment])

Average Interest Rate

Average Interest Rate =

AVERAGE(Financial_Loan[int_rate])

Average DTI

Average DTI =

AVERAGE(Financial_Loan[dti])

Good Loan Analysis Measures

Good Loan Applications

Good Loan Applications =

CALCULATE(
 DISTINCTCOUNT(Financial_Loan[id]),
 Financial_Loan[loan_status] IN {"Fully Paid","Current"}
)

Good Loan Percentage

Good Loan Percentage =

DIVIDE(
 [Good Loan Applications],
 [Total Loan Applications]
) * 100

Good Loan Funded Amount

Good Loan Funded Amount =

CALCULATE(
 SUM(Financial_Loan[funded_amount])
 WHERE Financial_Loan[loan_status] IN {"Fully Paid","Current"}
)

```
SUM(Financial_Loan[loan_amount]),  
Financial_Loan[loan_status] IN {"Fully Paid","Current"}  
)
```

Good Loan Amount Received

Good Loan Amount Received =

```
CALCULATE(  
SUM(Financial_Loan[total_payment]),  
Financial_Loan[loan_status] IN {"Fully Paid","Current"}  
)
```

Bad Loan Analysis Measures

Bad Loan Applications

Bad Loan Applications =

```
CALCULATE(  
DISTINCTCOUNT(Financial_Loan[id]),  
Financial_Loan[loan_status] = "Charged Off"  
)
```

Bad Loan Percentage

Bad Loan Percentage =

```
DIVIDE(  
[Bad Loan Applications],  
[Total Loan Applications]  
) * 100
```

Bad Loan Funded Amount

Bad Loan Funded Amount =
CALCULATE(
SUM(Financial_Loan[loan_amount]),
Financial_Loan[loan_status] = "Charged Off"
)

Bad Loan Amount Received

Bad Loan Amount Received =
CALCULATE(
SUM(Financial_Loan[total_payment]),
Financial_Loan[loan_status] = "Charged Off"
)

Loan Term Analysis Measures

36-Month Loan Applications

36-Month Loan Applications =
CALCULATE(
DISTINCTCOUNT(Financial_Loan[id]),
Financial_Loan[term] = "36 months"
)

60-Month Loan Applications

60-Month Loan Applications =
CALCULATE(
DISTINCTCOUNT(Financial_Loan[id]),
Financial_Loan[term] = "60 months"
)

Average Loan Amount

Average Loan Amount =

AVERAGE(Financial_Loan[loan_amount])

Average Annual Income

Average Annual Income =

AVERAGE(Financial_Loan[annual_income])

Dashboard Pages

Page 1 – Good Loan vs Bad Loan Analysis

Purpose:

- Analyze loan performance based on loan status.
- Compare Good Loans and Bad Loans.

KPIs:

- Good Loan Percentage
- Good Loan Applications
- Bad Loan Percentage
- Bad Loan Applications

Visuals:

- Good vs Bad Loan Applications
- Funded Amount by Loan Status
- Total Amount Received by Loan Status
- Loan Status Distribution

Page 2 – Monthly Trend and Regional Analysis

Purpose:

- Monitor loan trends over time.
- Analyze lending activity across different states.

KPIs:

- Total Applications
- Total Funded Amount
- Total Amount Received
- Average Interest Rate

Visuals:

- Monthly Loan Application Trend
- Monthly Funded Amount Trend
- State-wise Loan Applications Map
- Top States by Funded Amount

Page 3 – Loan Term and Employment Length Analysis

Purpose:

- Understand borrower repayment preferences.
- Analyze loan applications across employment categories.

KPIs:

- 36-Month Loan Applications
- 60-Month Loan Applications
- Average Loan Amount
- Average Annual Income

Visuals:

- Loan Distribution by Term
- Funded Amount by Loan Term
- Applications by Employment Length
- Funded Amount by Employment Length

Page 4 – Loan Purpose and Home Ownership Analysis

Purpose:

- Identify major borrowing reasons.
- Compare lending patterns across ownership categories.

KPIs:

- Total Applications
- Total Funded Amount
- Average Interest Rate
- Average DTI

Visuals:

- Loan Applications by Purpose
- Funded Amount by Purpose
- Applications by Home Ownership Status
- Lending Patterns by Home Ownership Category

Interactive Features

Bookmarks Implemented

- Default Dashboard View
- Good Loan View
- Bad Loan View
- KPI View 1
- KPI View 2

Action Buttons Implemented

- Home Navigation
- Back Navigation
- Next Navigation
- Bookmark Navigation
- KPI Switching Buttons

These features improve user experience and provide interactive report exploration.

Business Benefits

- Enables monitoring of loan portfolio health.
- Helps identify high-risk loan segments.
- Supports data-driven lending decisions.
- Improves visibility into borrower behavior.
- Provides quick access to key performance indicators.
- Facilitates management reporting through interactive dashboards.

Conclusion

- The Bank Loan Analysis Dashboard successfully transforms raw loan data into meaningful business insights.
- Through data cleaning, DAX calculations, interactive visualizations, bookmarks, and navigation features, the solution provides a comprehensive view of loan performance and borrower trends, supporting informed business decision-making.



